



Toward Self-Evolving AI Ecosystems



Observe → Synthesize → Collaborate → Evolve

Stage I: Trace Capture

Capture execution evidence from real-world operations.

Performing Tasks

- ✓ Data Processing
- ✓ Model Inference
- ✓ Tool Use
- ✓ Error Handling
- ✓ Report Generation
- ✓ ...



TRACE CAPTURE UNIT

Execution Traces

Log Details

[2025-0-08 10:15:23] Start job
[2025-0-08 10:16:47] Fetch data
[2025-0-08 10:17:02] Invoke API
[2025-0-08 10:18:05] Timeout
[2025-0-08 10:20:10] Retry....

Task Replays

1. Data query replay ✓
2. Model run replay ✓
3. Tool call replay ✗
4. ...

State Snapshots

```
{  
  "state_id": "tx_10093",  
  "context": "user_query",  
  "intermediate": [...]  
}
```

Capturing operational evidence for learning and improvement.

Stage II: Beyond-Gradient Synthesis

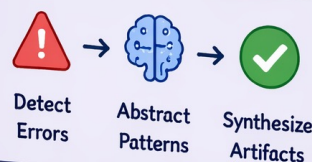
Convert traces into knowledge artifacts—errors become tests, successes become skills.

Analyzing traces...
Synthesizing beyond-gradient knowledge.



Execution Traces

BEYOND-GRADIENT SYNTHESIZER



Errors → Regression Tests

- ✓ Reproduce failure
- ✓ Minimize example
- ✓ Create test case
- ✓ Prevent recurrence

Successes → Reusable Skills

- ✓ Generalize pattern
- ✓ Document procedure
- ✓ Package skill module
- ✓ Enable reuse

Learning without gradient. Improving without forgetting.

Stage III: Colleague Collaboration

Validate, deploy, and evolve together as digital colleagues.

Testing new skill in the sandbox...



Validation Checklist

- ✓ Functional Correctness
- ✓ Edge Case Handling
- ✓ Safety & Security
- ✓ Performance Check
- ✓ Documentation
- ✓ ...

Validation Successful!

Great update!
Ship it! 🚀

SYSTEM DASHBOARD

- #### System Updates
- ✓ New Skills Deployed
 - ✓ Tests Passed
 - ✓ Performance Improved
 - ✓ Security Verified



Agreed!
Deploying! ✓

Shared Goals

Transparent Communication

Continuous Feedback

Co-Evolution